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6 In animal cells the nucleus can divide by mitosis or meiosis.

(a) In the cell cycle, the time between two divisions of mitosis is called interphase.

(i) Describe what happens to DNA during interphase.

(2)

(ii) Name the stage of mitosis that follows prophase.

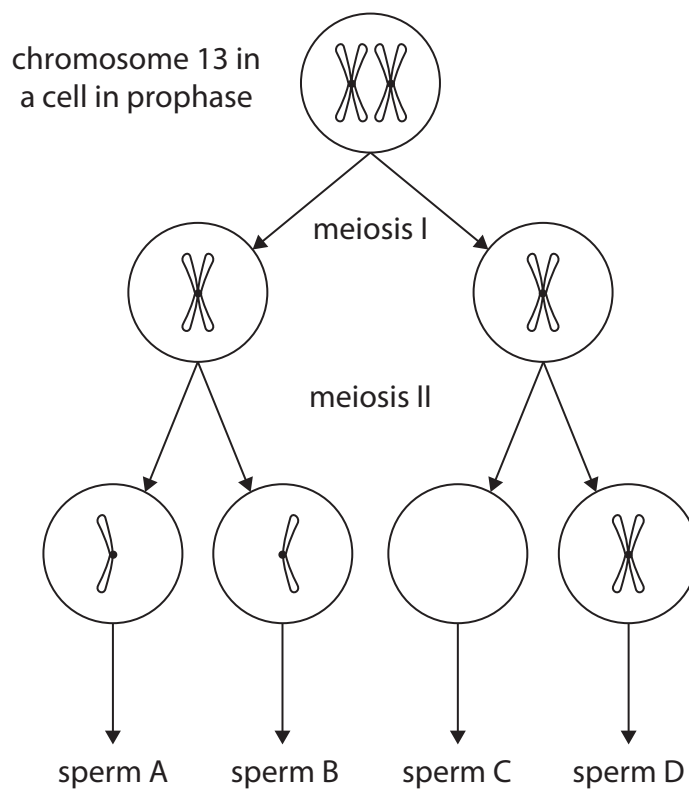
(1)



- (b) A person with Patau syndrome had three copies of chromosome 13 in each body cell.

This occurred due to an error in meiosis in the formation of his father's sperm cells.

The diagram shows what happened to chromosome 13 during the formation of the father's sperm cells.



- (i) A zygote is formed at fertilisation.

Explain how an error in meiosis could result in the production of a zygote with three copies of chromosome 13.

(4)

- (ii) The zygote would develop into a morula and then into a blastocyst.

Which row in the table is correct?

(1)

	the zygote is:	cells in the blastocyst inner mass are:
☐ A	pluripotent	pluripotent
☐ B	pluripotent	totipotent
☐ C	totipotent	pluripotent
☐ D	totipotent	totipotent

- (c) An individual can also develop Patau syndrome if part of chromosome 13 becomes attached to a different chromosome during prophase I.

Suggest a reason why part of chromosome 13 can become attached to a different chromosome when a gamete is formed.

(1)

- (d) Methylation of DNA can alter the expression of certain genes on chromosome 13.

Methylation of DNA can be passed on during mitotic cell division.

- (i) Explain how DNA methylation can alter the expression of some genes.

(2)

- (ii) Explain why epigenetic changes to genes, such as DNA methylation, are passed on by mitosis.

(2)

(Total for Question 6 = 13 marks)

